SSRD Project: Predicting Sleep Quality

2025-04-29

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```r
str(se)
```

```
## 'data.frame':    452 obs. of  15 variables:
##  $ ID                    : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Age                   : int  65 69 40 40 57 36 27 53 41 11 ...
##  $ Gender                : chr  "Female" "Male" "Female" "Female" ...
##  $ Bedtime               : chr  "2021-03-06 01:00:00" "2021-12-05 02:00:00" "2021-05-25 21:30:00" "2(
##  $ Wakeup.time           : chr  "2021-03-06 07:00:00" "2021-12-05 09:00:00" "2021-05-25 05:30:00" "2(
##  $ Sleep.duration        : num  6 7 8 6 8 7.5 6 10 6 9 ...
##  $ Sleep.efficiency      : num  0.88 0.66 0.89 0.51 0.76 0.9 0.54 0.9 0.79 0.55 ...
##  $ REM.sleep.percentage  : int  18 19 20 23 27 23 28 28 28 18 ...
##  $ Deep.sleep.percentage : int  70 28 70 25 55 60 25 52 55 37 ...
##  $ Light.sleep.percentage: int  12 53 10 52 18 17 47 20 17 45 ...
##  $ Awakenings            : num  0 3 1 3 3 0 2 0 3 4 ...
##  $ Caffeine.consumption  : num  0 0 0 50 0 NA 50 50 50 0 ...
##  $ Alcohol.consumption   : num  0 3 0 5 3 0 0 0 0 0 ...
##  $ Smoking.status        : chr  "Yes" "Yes" "No" "Yes" ...
##  $ Exercise.frequency    : num  3 3 3 1 3 1 1 3 1 0 ...
```

```r
wse = dplyr::select(se, c(Deep.sleep.percentage,
                  REM.sleep.percentage,
                  Sleep.efficiency,
                  Caffeine.consumption,
                  Awakenings,
                  Exercise.frequency,
                  Alcohol.consumption,
                  Gender,
                  Smoking.status))

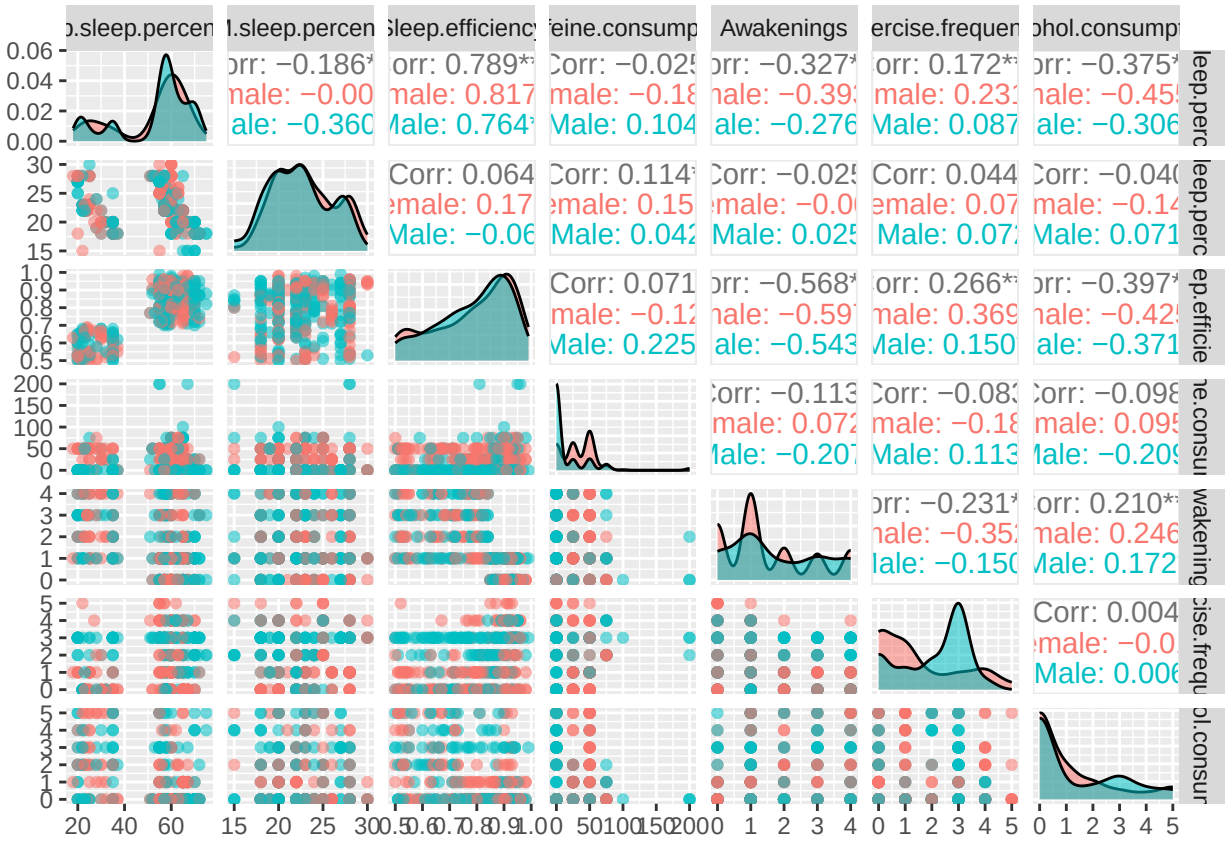
wse = drop_na(wse)
```

```r
count(filter(wse, Gender == "Male"))
```

```
##     n
## 1 194
```

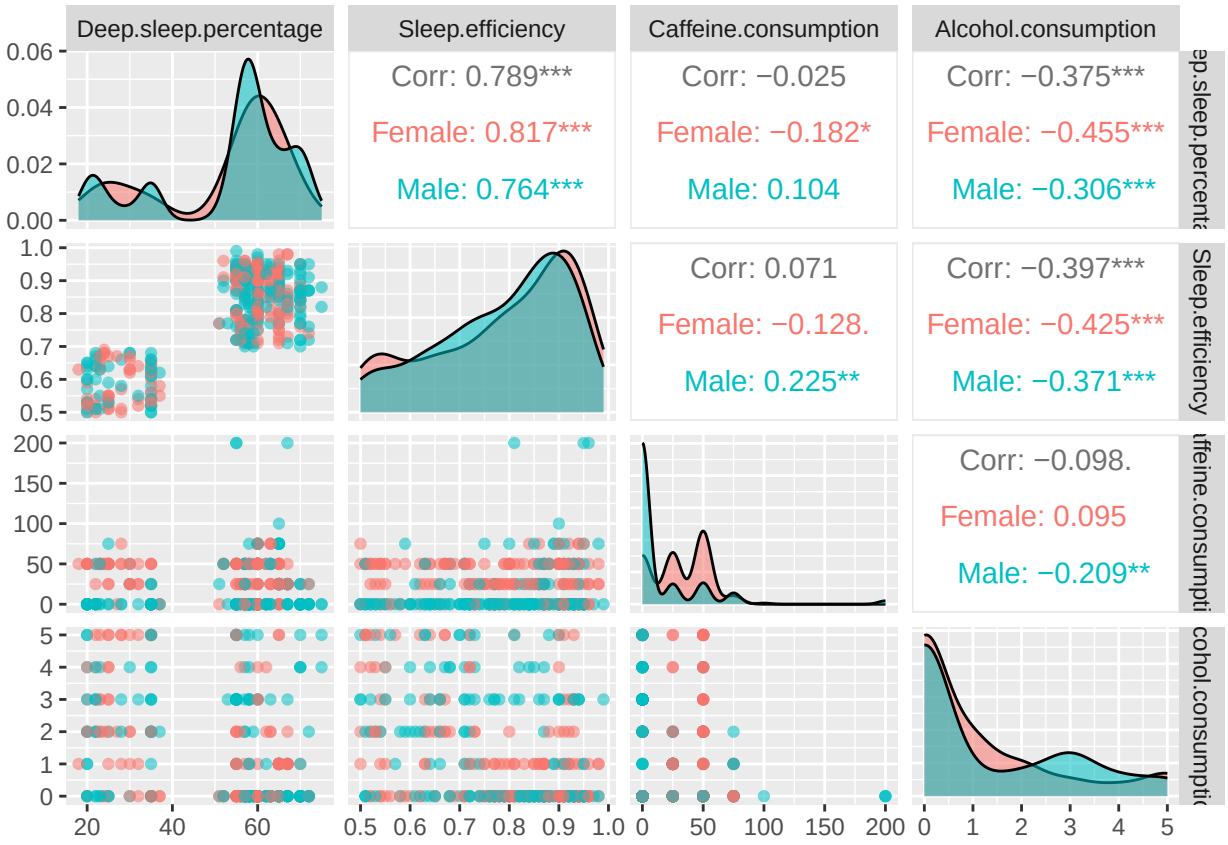
GG PAIR PLOTS

All variables by gender.

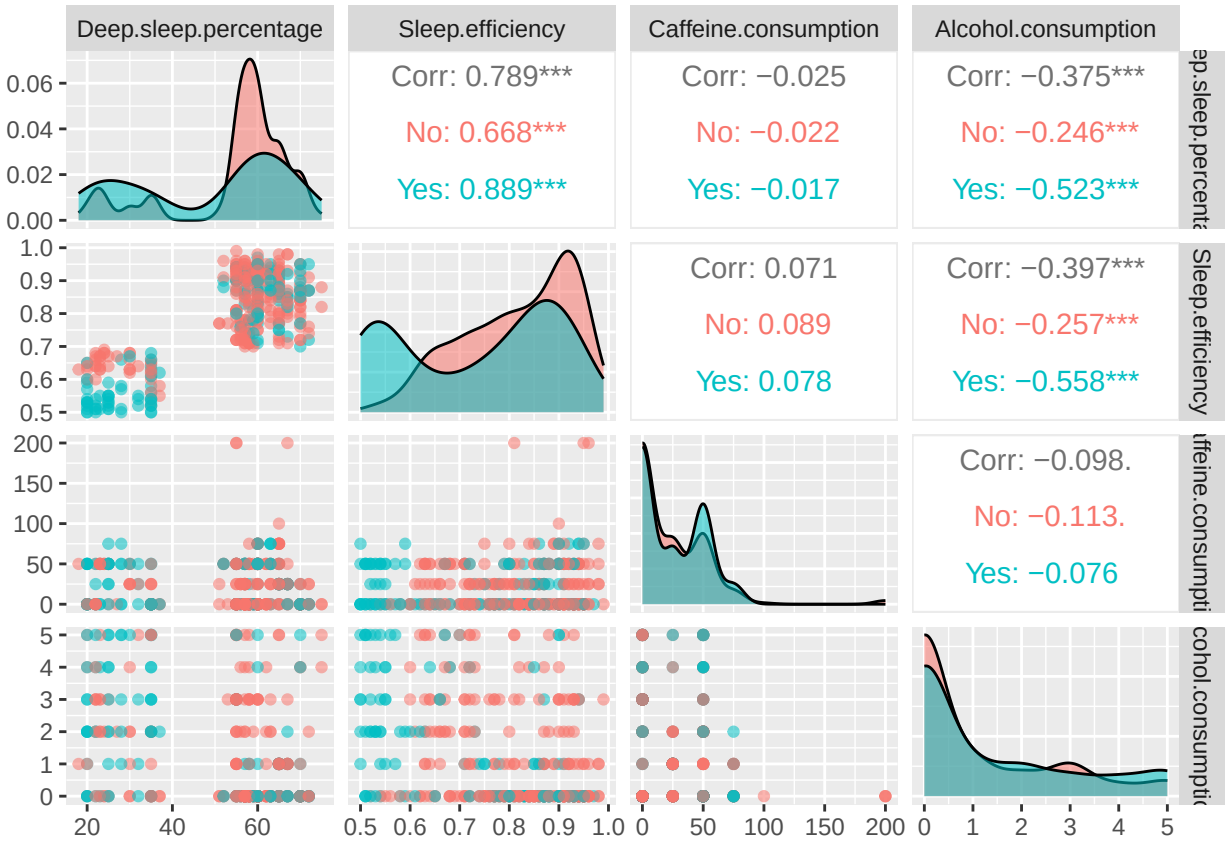


Looking at this pairs plot, we found that there were situations where gender was a mediator between our sleep variables and substance use. For example, Caffeine consumption is not significantly associated with deep sleep. However, grouping by gender reveals caffeine consumption negatively predicts deep sleep percentage for females. With respect to healthy habits, exercise frequency was positively associated with deep sleep for the full sample, but this effect was driven by the strength of significance for females only. For our analysis, we explored the effects of alcohol consumption and caffeine consumption on deep sleep and sleep efficiency. We chose alcohol and caffeine because of their differing effects as a depressant and stimulant, respectively. We then explore the strength of these effects on smokers and non-smokers on awakenings and sleep efficiency. All effects were analyzed with respect to gender.

Select variables by gender

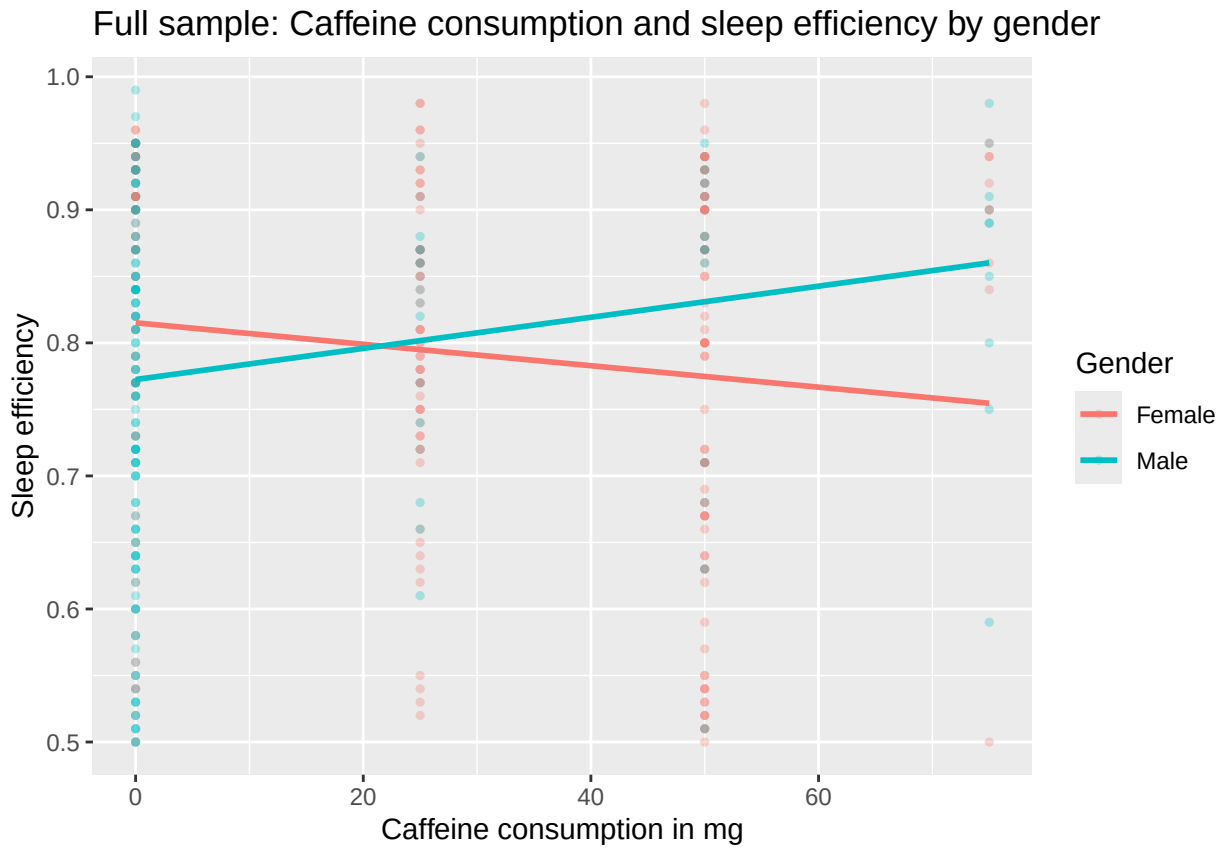


Select variables by smoking status



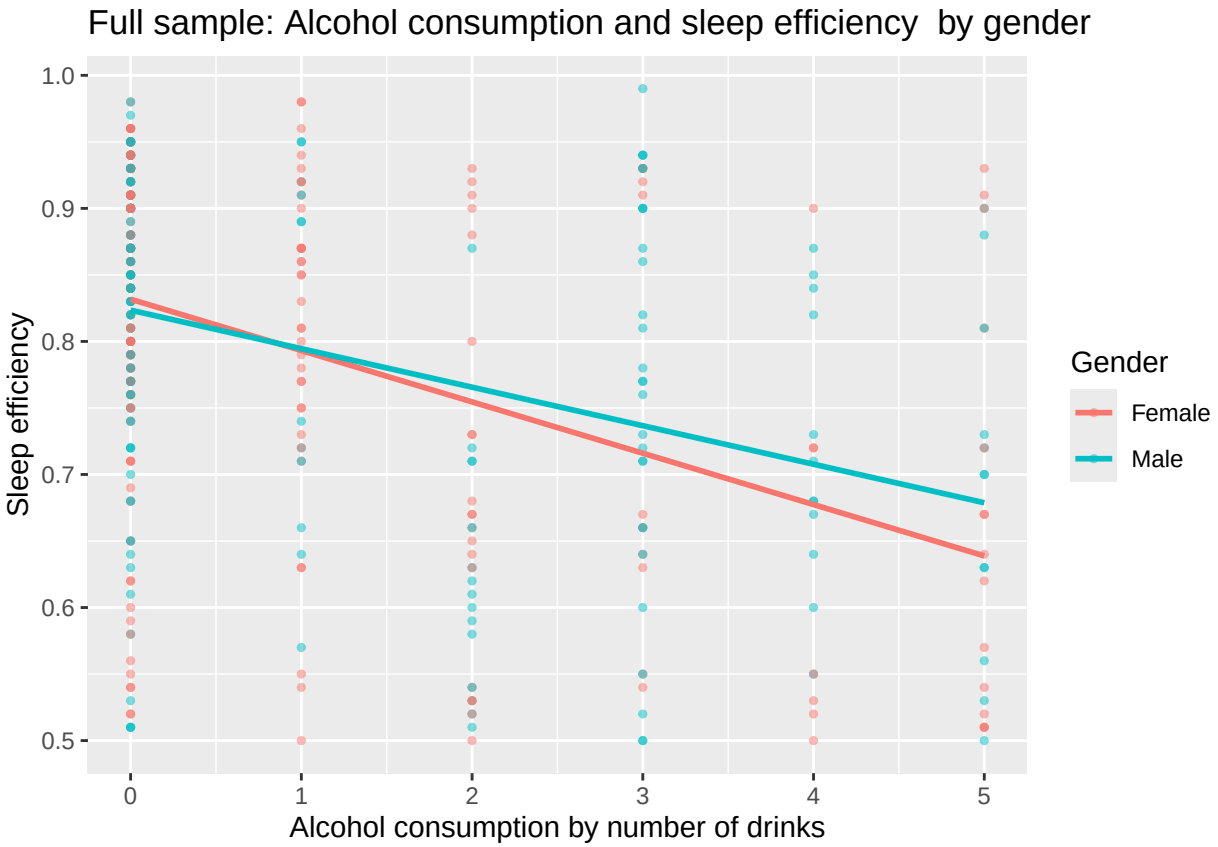
Linear Models

Plot: Caffeine consumption and sleep efficiency by gender (full sample)



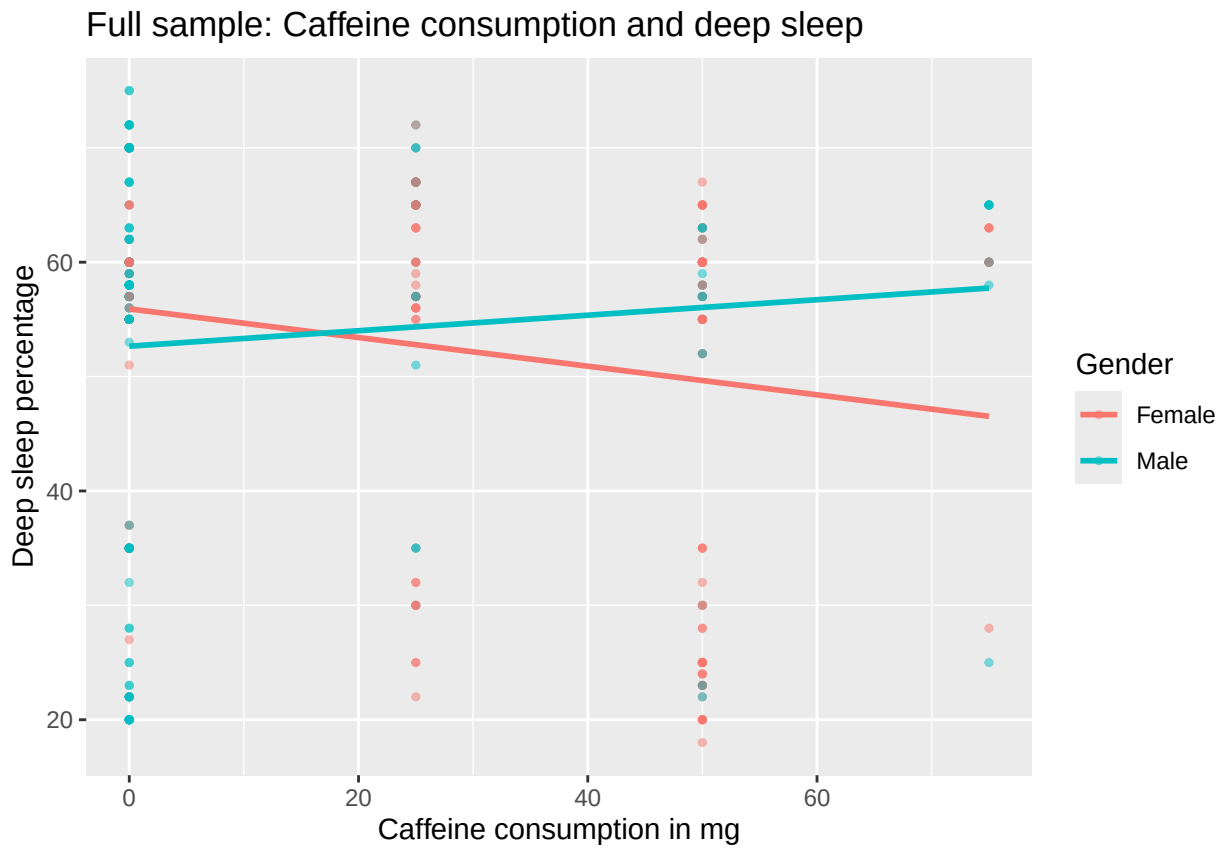
For the full sample, caffeine consumption and sleep efficiency were significantly and positively associated for males, and negatively but NOT significantly associated for females. This suggests that males are more likely to sleep more efficiently as they consumer more caffeine, up to 80mg.

Plot: Alcohol consumption and sleep efficiency by gender (full sample)



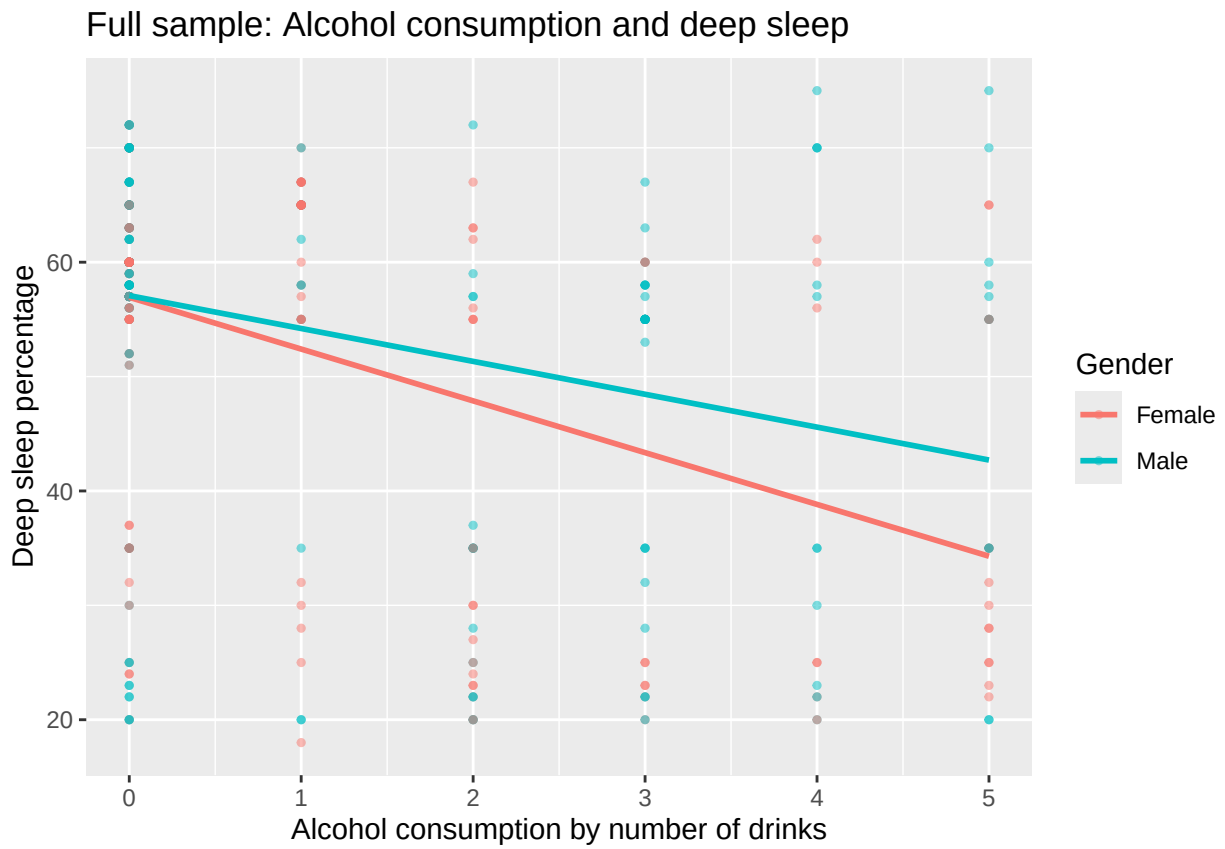
For the full sample, alcohol consumption and sleep efficiency were significantly and negatively associated for both males and females, with a slightly stronger effect for females at around 5 drinks (19% reduction for females compared to 16% for males).

Plot: Caffeine consumption and deep sleep (full sample)



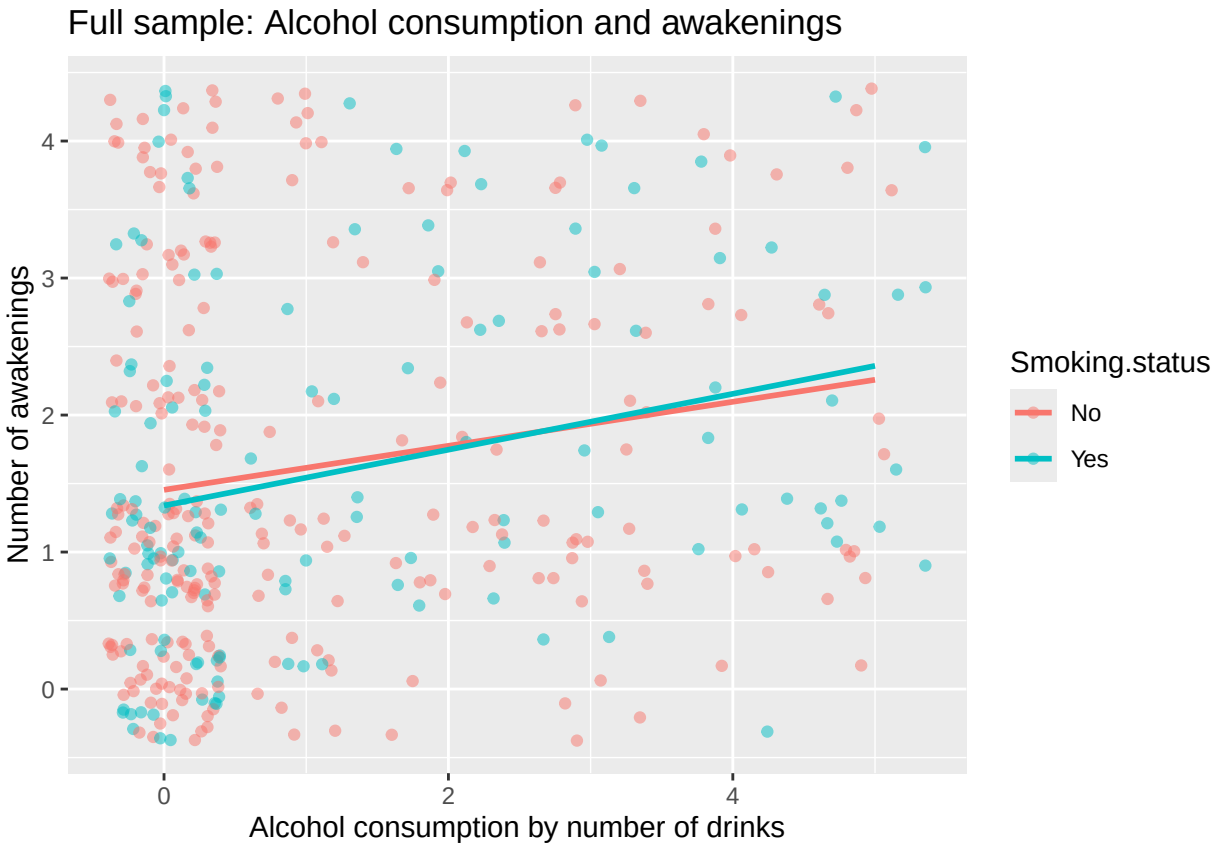
For the full sample, caffeine consumption was significantly and positively associated for males, but significantly and negatively associate for females. This is consistent with the findings for sleep efficiency. However, a portion of data is missing for deep sleep percentage (from 40 to 50%). As a results, the finding here is less reliable and should be corrected in future studies.

Plot: Alcohol consumption and deep sleep (full sample)



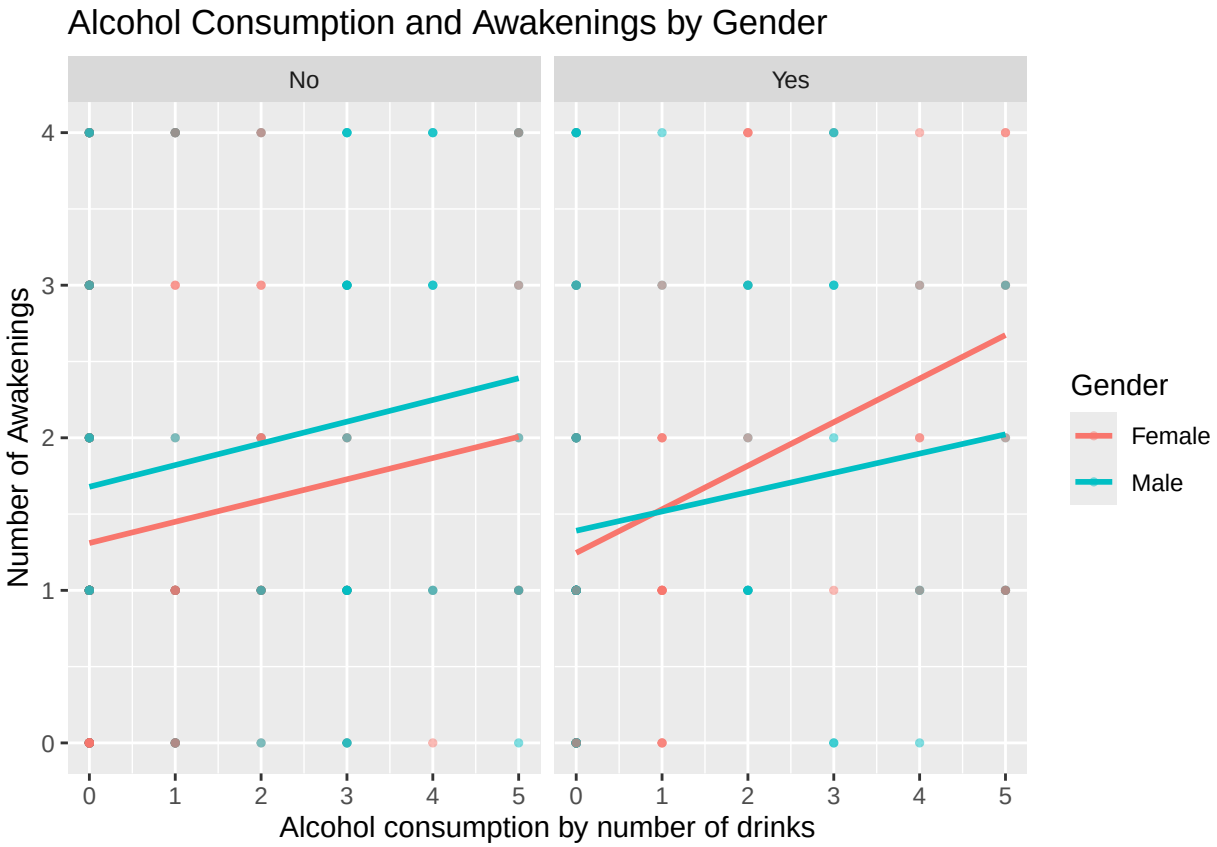
For the full sample, alcohol consumption and deep sleep were significantly and negatively associated for both males and females, with a slightly stronger effect for females. Here again, however, missing data calls this result into question. Future studies should take this into consideration.

Plot: Alcohol consumption and awakenings (full sample)



In the full sample, alcohol consumption and awakenings were significantly and postively associated for both males and females. From 0 to 5 drinks, the effect is about the same for both males and females.

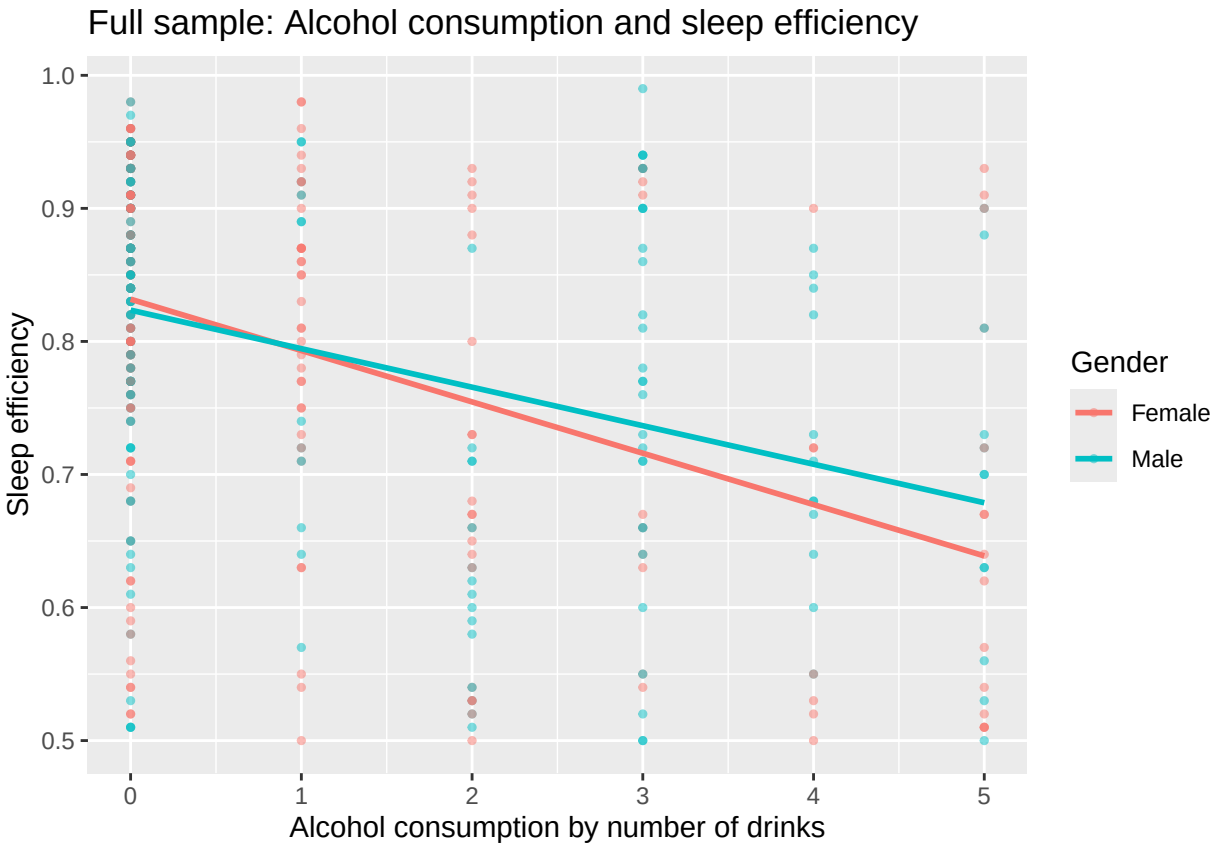
Plot: Alcohol consumption and awakenings by gender and smoking status



Filtered for non-smokers, alcohol consumption and awakenings were positively associated, but NOT significant.

Filtered for smokers, alcohol consumption and awakenings were significantly and positively associated for both males and females, with a stronger effect on females.

Plot: Alcohol consumption and sleep efficiency (full sample)



For the sample, alcohol consumption and sleep efficiency are significantly and negatively associated for males and females, with a slightly strong effect on females at around 5 drinks. From 0 to 5 drinks, there is about a 15% reduction for males and a 20% reduction for females.

LM: Interaction of alcohol and gender on sleep efficiency

```
mod3 = lm(Sleep.efficiency ~
           Alcohol.consumption *
             Gender +
             0,
         data = wse)

summary(mod3)
```

```
##
## Call:
## lm(formula = Sleep.efficiency ~ Alcohol.consumption * Gender +
##     0, data = wse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.31351 -0.07456  0.01827  0.09649  0.29119
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## Alcohol.consumption           -0.038583   0.005762  -6.696 7.64e-11 ***
## GenderFemale                   0.831728   0.010883  76.425  < 2e-16 ***
## GenderMale                     0.823507   0.011121  74.051  < 2e-16 ***
## Alcohol.consumption:GenderMale  0.009636   0.007892   1.221    0.223
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1248 on 384 degrees of freedom
## Multiple R-squared:  0.976,  Adjusted R-squared:  0.9757
## F-statistic:  3899 on 4 and 384 DF,  p-value: < 2.2e-16
```

LM: Interaction of alcohol and smoking on sleep efficiency

```
mod4a = lm(Sleep.efficiency ~
           Alcohol.consumption *
             Smoking.status +
             0,
         data = wse)

summary(mod4a)
```

```
##
## Call:
## lm(formula = Sleep.efficiency ~ Alcohol.consumption * Smoking.status +
##     0, data = wse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.31752 -0.07752  0.01248  0.09248  0.35394
##
## Coefficients:
##                                        Estimate Std. Error t value Pr(>|t|)
## Alcohol.consumption                   -0.018575   0.004780  -3.886  0.00012 ***
## Smoking.statusNo                       0.837517   0.008895  94.157  < 2e-16 ***
## Smoking.statusYes                      0.802033   0.012756  62.874  < 2e-16 ***
## Alcohol.consumption:Smoking.statusYes -0.032620   0.007549  -4.321 1.98e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.117 on 384 degrees of freedom
## Multiple R-squared:  0.9789, Adjusted R-squared:  0.9786
## F-statistic:  4447 on 4 and 384 DF,  p-value: < 2.2e-16
```

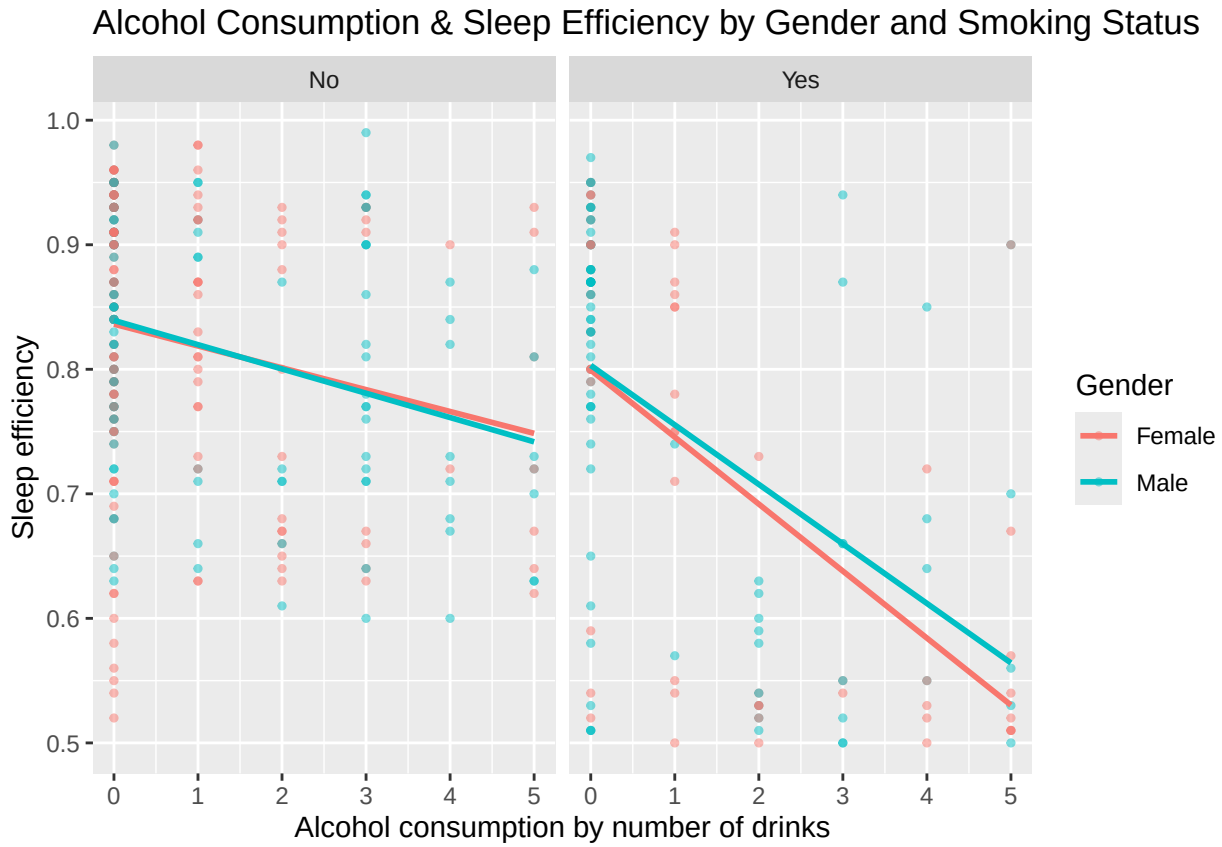
LM: Interaction of alcohol and smoking on deep sleep

```
mod4b = lm(Deep.sleep.percentage ~
           Alcohol.consumption *
             Smoking.status +
             0,
         data = wse)

summary(mod4b)
```

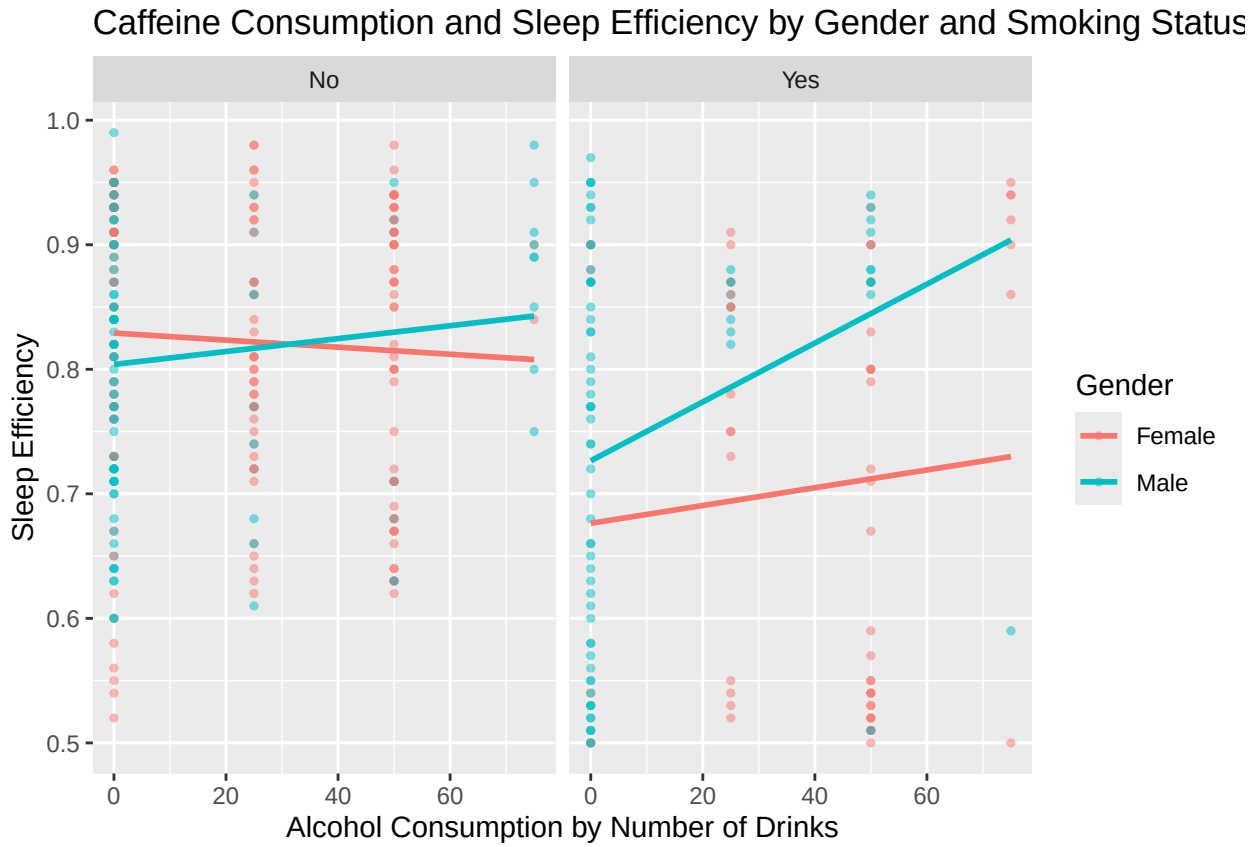
```
##
## Call:
## lm(formula = Deep.sleep.percentage ~ Alcohol.consumption * Smoking.status +
##     0, data = wse)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -37.504  -2.504   2.496   9.004  41.723
##
## Coefficients:
##                                       Estimate Std. Error t value Pr(>|t|)
## Alcohol.consumption                    -2.1695     0.5707  -3.802 0.000167 ***
## Smoking.statusNo                       57.5043     1.0619  54.150  < 2e-16 ***
## Smoking.statusYes                      55.4156     1.5229  36.387  < 2e-16 ***
## Alcohol.consumption:Smoking.statusYes  -3.2582     0.9012  -3.615 0.000340 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.97 on 384 degrees of freedom
## Multiple R-squared:  0.9363, Adjusted R-squared:  0.9356
## F-statistic:  1411 on 4 and 384 DF,  p-value: < 2.2e-16
```

Plot: Alcohol Consumption & Sleep Efficiency by Gender and Smoking Status



Filtered for non-smokers, alcohol consumption and sleep efficiency are still significantly and negatively associated. However, the difference between males and females at 5 drinks is much less (from 0 to 5 drinks, about an 8% decrease for males, and 10% decrease for females), indicating the effect must be stronger for smokers.

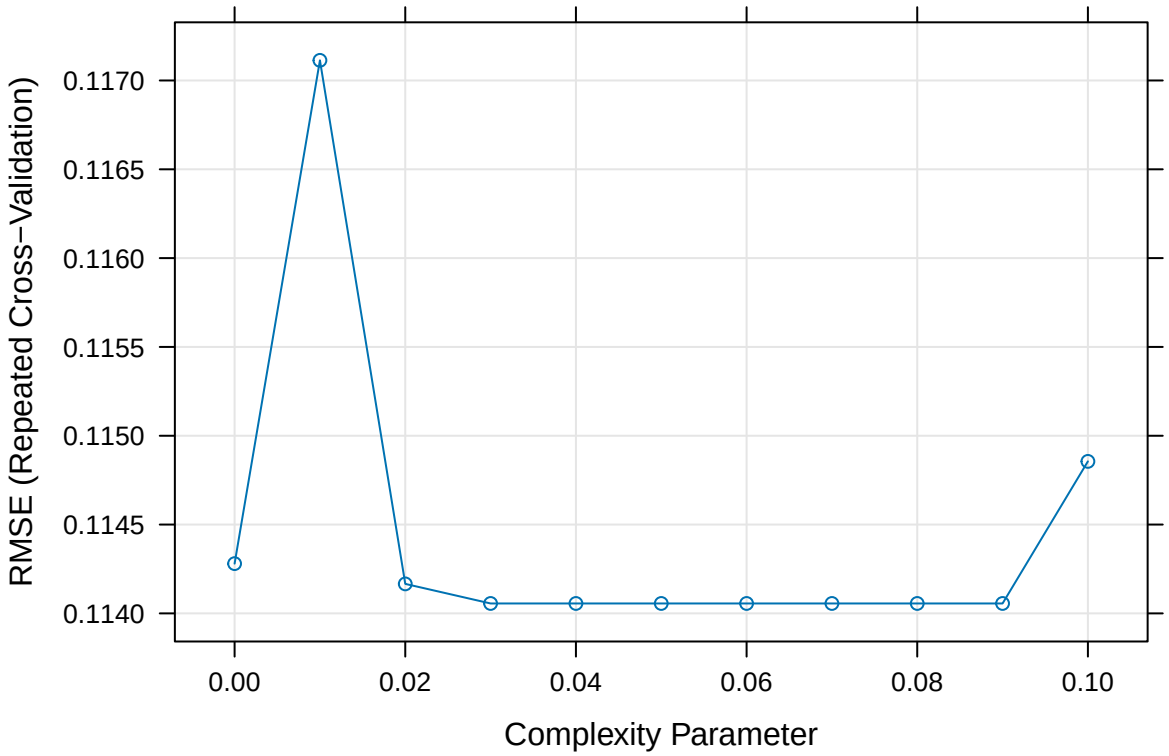
Plot: Caffeine Consumption and Sleep Efficiency by Gender and Smoking Status

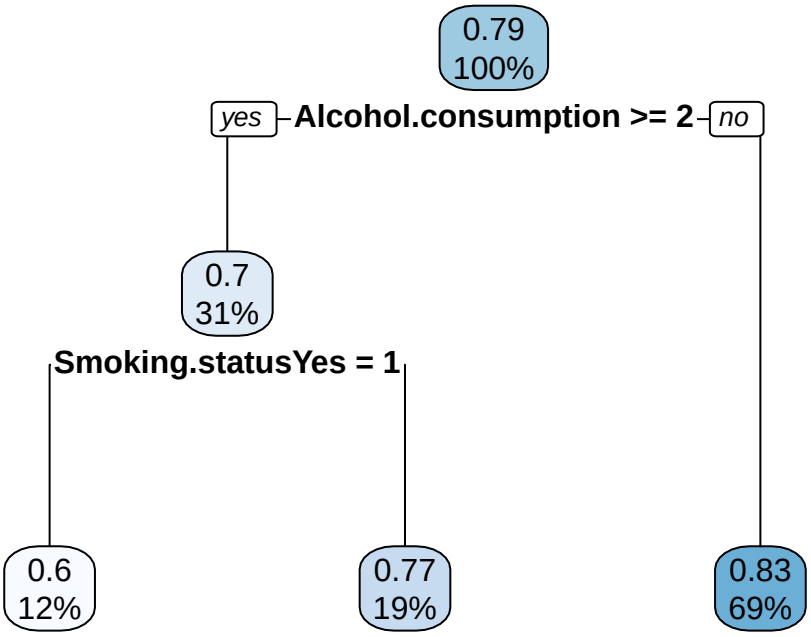


When the sample is filtered for smokers only, the effect of alcohol consumption is significant and much more negative. From 0 to 5 drinks, there is about a 30% decrease in sleep efficiency compared to about 6 to 7% for non-smokers.

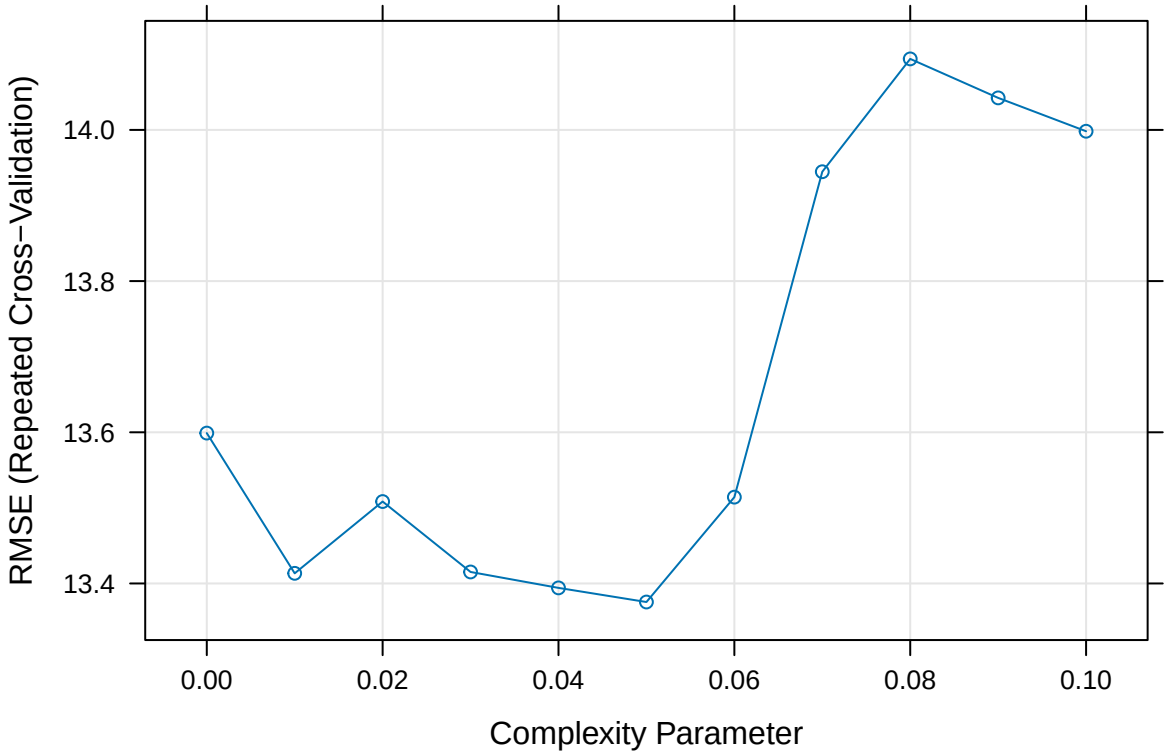
Tree Models

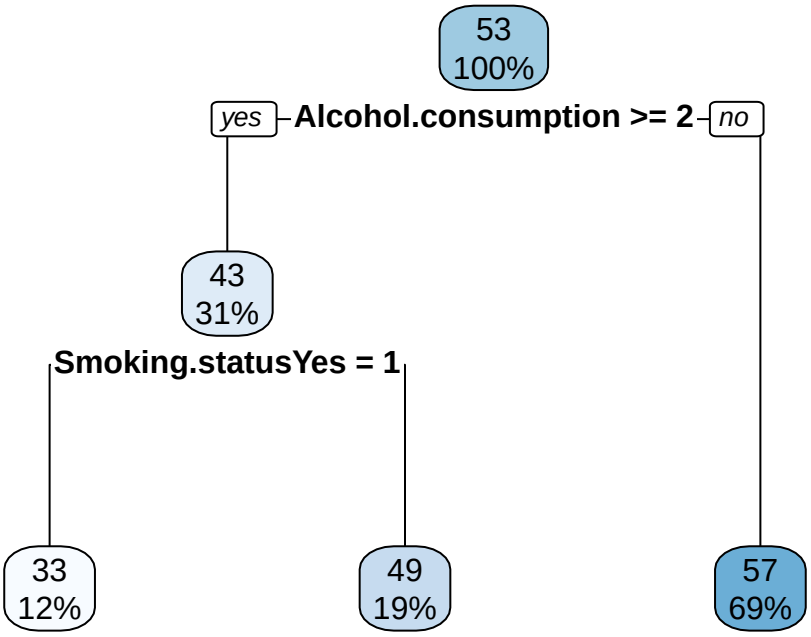
Sleep efficiency




0.79
100%
yes
Alcohol.consumption >= 2
no
0.7
31%
Smoking.statusYes = 1
0.6
12%
0.77
19%
0.83
69%

Deep sleep


RMSE (Repeated Cross−Validation)
14.0
13.8
13.6
13.4
0.00
0.02
0.04
0.06
0.08
0.10
Complexity Parameter


53
100%
yes
Alcohol.consumption >= 2
no
43
31%
Smoking.statusYes = 1
33
12%
49
19%
57
69%

Prepare data for LDA & QDA

Modify Data

.

```
# Create outcome variable with two levels
# 0 < Poor sleep efficiency <= 0.8
# 80 < Good sleep efficiency < 100
wse = wse %>% mutate(y = Sleep.efficiency > .8) # sleep efficiency threshold
wse$y = as.factor(as.integer(as.logical(wse$y)))
```

```
# To prevent partimat plot errors
qda_data = dplyr::select(wse,
                         Caffeine.consumption,
                         Exercise.frequency,
                         Alcohol.consumption,
                         Smoking.status,
                         y)
```

Train and Test

```
set.seed(85)
sampleInd = sample(1:388, 272)
wse.train = qda_data[sampleInd,]
wse.test = qda_data[-sampleInd,]
```

Model: LDA

Step 1: Fit LDA

.

```
# Fit the model
lda.wse = lda(y~., data = wse.train)
lda.wse
```

```
## Call:
## lda(y ~ ., data = wse.train)
##
## Prior probabilities of groups:
##         0         1
## 0.4338235 0.5661765
##
## Group means:
##   Caffeine.consumption Exercise.frequency Alcohol.consumption Smoking.statusYes
## 0             21.82203           1.262712           1.6525424         0.4406780
## 1             25.81169           2.136364           0.7337662         0.2792208
##
## Coefficients of linear discriminants:
##                                LD1
## Caffeine.consumption  0.005636744
## Exercise.frequency    0.524619459
## Alcohol.consumption  -0.424845242
## Smoking.statusYes    -0.741097821
```

Step 2: Prediction

.

```
# Make predictions
predLDA = lda.wse %>% predict(wse.test)

# Predicted classes
head(predLDA $class, 6)
```

```
## [1] 0 1 1 1 1 1
## Levels: 0 1
```

```
# Predicted probabilities of class memebership.
head(predLDA $posterior, 6)
```

```
##            0         1
## 2  0.6038016 0.3961984
## 5  0.4223921 0.5776079
## 8  0.3067329 0.6932671
## 10 0.2458723 0.7541277
## 12 0.3067329 0.6932671
## 13 0.1713922 0.8286078
```

```
# Linear discriminants
head(predLDA $x, 3)
```

```
##           LD1
## 2 -0.75954041
## 5 -0.01844259
## 8  0.48869140
```

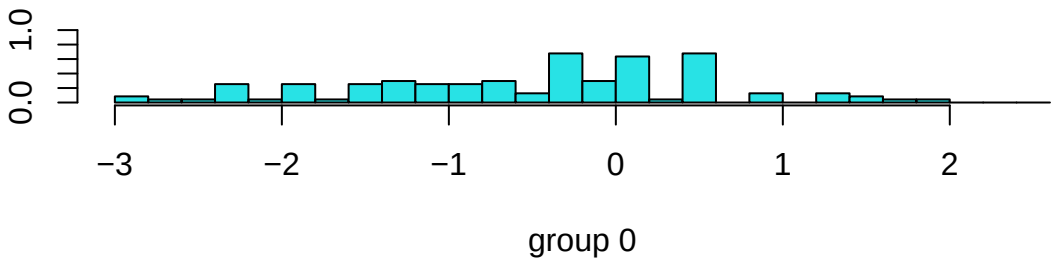
Step 3: Model Accuracy

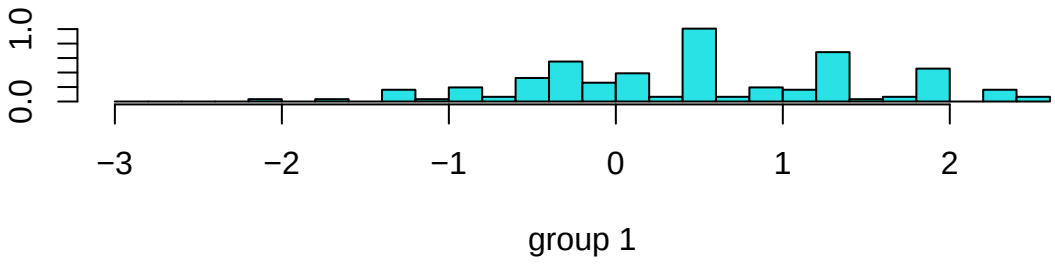
.

```
# Model accuracy
mean(predLDA$class==wse.test$y)
```

```
## [1] 0.6724138
```

```
plot(lda.wse)
```

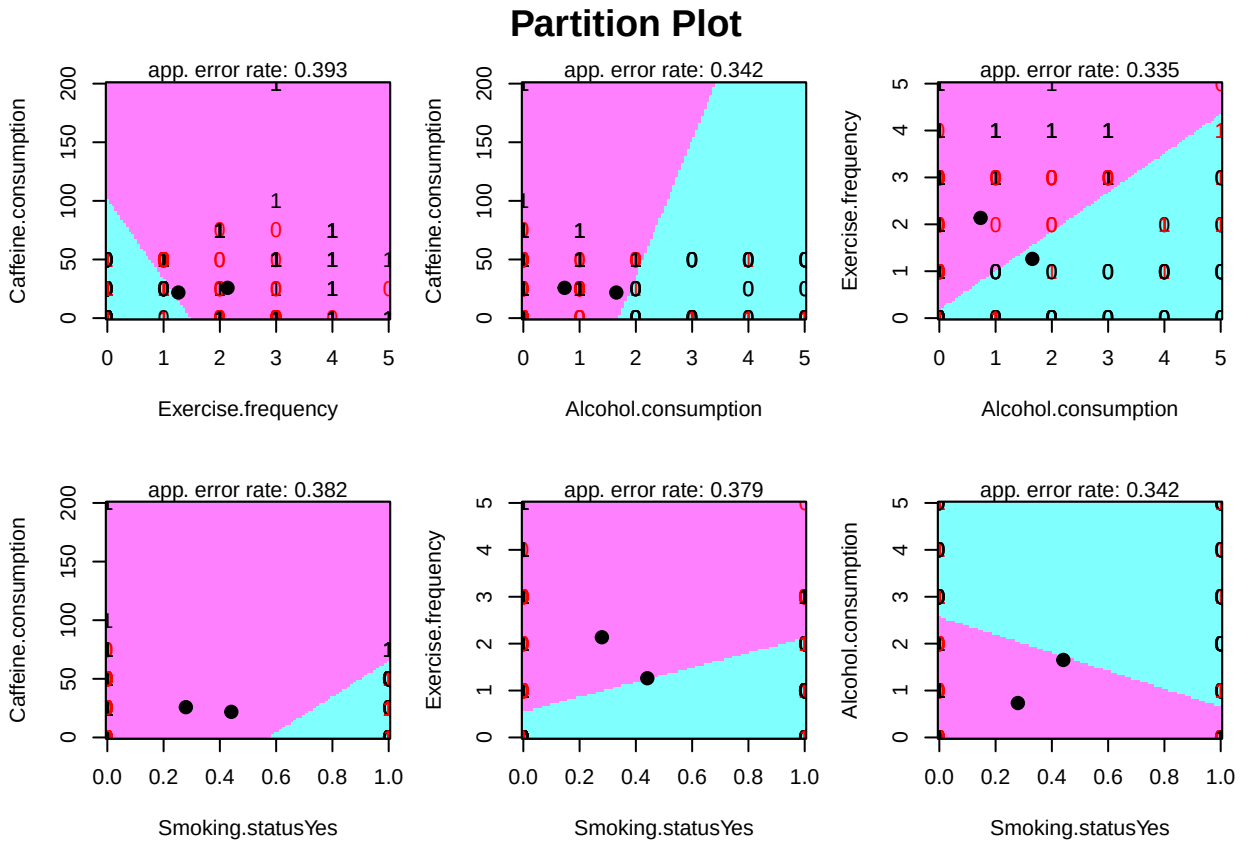




Step 4: Boundary Line

.

```
partimat(y~., data = wse.train, method = "lda")
```



Model: QDA

Step 1: Fit a QDA

```
# Fit the model
qda.wse <- qda(y~., data = wse.train)
qda.wse
```

```
## Call:
## qda(y ~ ., data = wse.train)
##
## Prior probabilities of groups:
##         0         1
## 0.4338235 0.5661765
##
## Group means:
##   Caffeine.consumption Exercise.frequency Alcohol.consumption Smoking.statusYes
## 0             21.82203           1.262712           1.6525424         0.4406780
## 1             25.81169           2.136364           0.7337662         0.2792208
```

Step 2: Predict

```
# Make predictions
predQDA <- qda.wse %>% predict(wse.test)
```

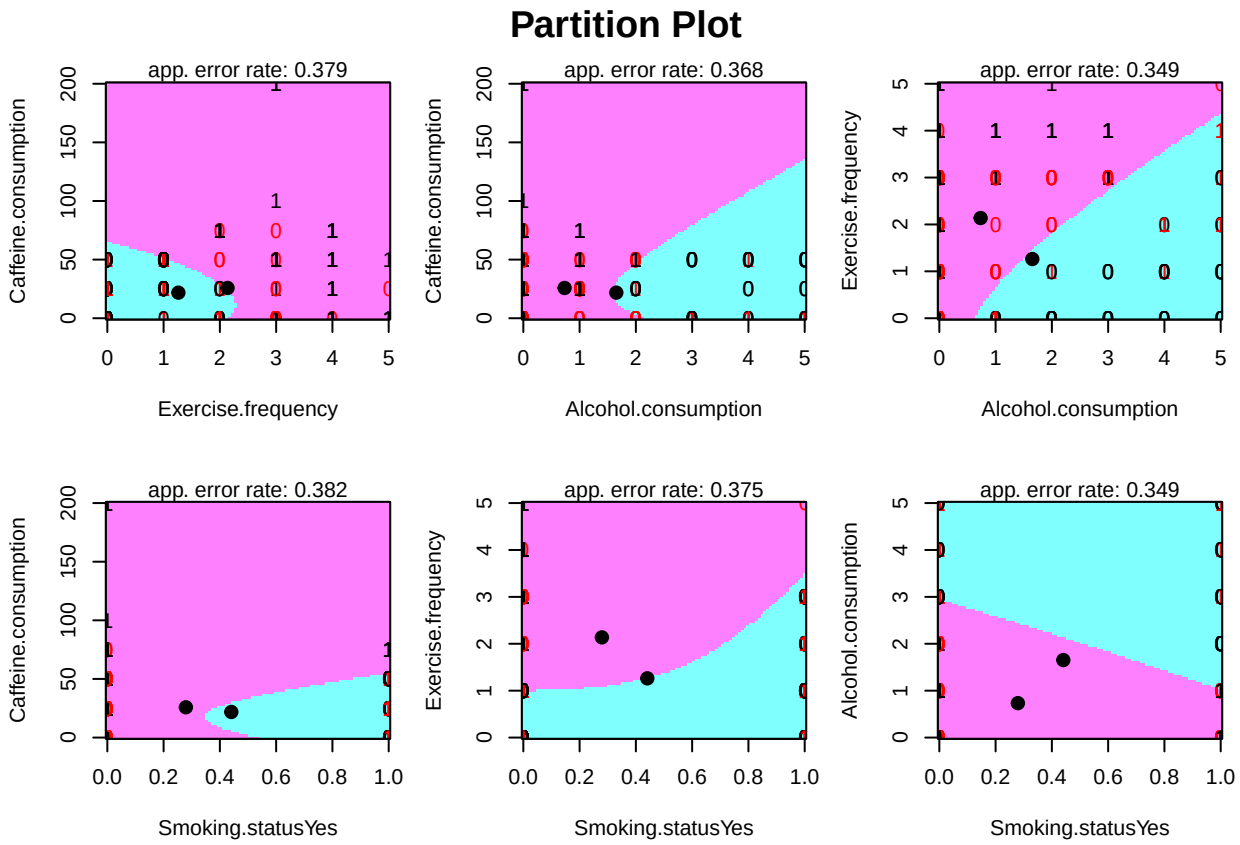
Step 3: Model Accuracy

```
# Model accuracy
mean(predQDA$class==wse.test$y)
```

```
## [1] 0.6724138
```

Step 4: Boundary Line

```
partimat(y~., data = wse.train, method = "qda")
```



Ridge Regression

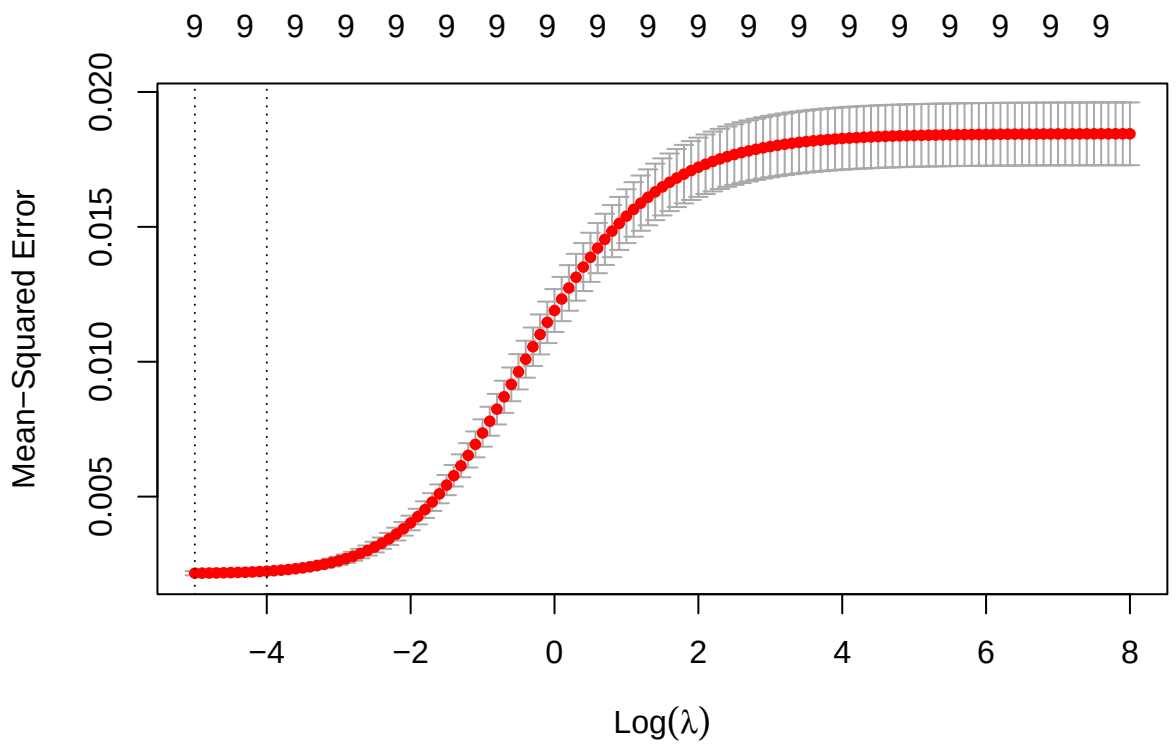
```
set.seed(50)

#RIDGE REGRESSION

#we need to define the model equation
X <- model.matrix(Sleep.efficiency ~., data=wse)[,-1]
#and the outcome
Y <- wse[,"Sleep.efficiency"]

#Penalty type (alpha=0 is ridge)
cv.lambda <- cv.glmnet(x=X, y=Y,
                       alpha = 0,
                       lambda=exp(seq(-5,8,.1)))
```

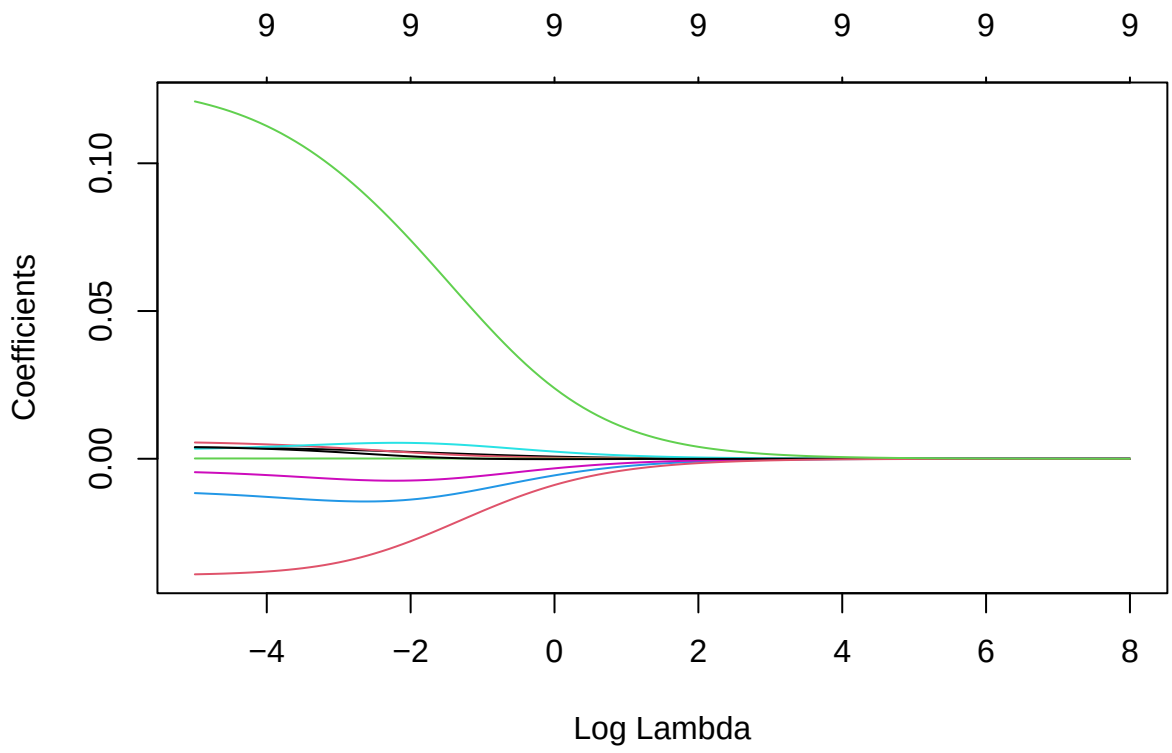
```
plot(cv.lambda)                                    #MSE for several lambdas
```




```
cv.lambda$lambda.min
```

```
## [1] 0.006737947
```

```r
#ridge path
plot(cv.lambda$glmnet.fit,
     "lambda", label=FALSE)
```



Final Model

```r
lmin        <- cv.lambda$lambda.min
ridge.model <- glmnet(x=X, y=Y,
                      alpha = 0,
                      lambda = lmin)
ridge.model$beta
```

```
## 9 x 1 sparse Matrix of class "dgCMatrix"
##                                    s0
## Deep.sleep.percentage  3.800573e-03
## REM.sleep.percentage   5.504615e-03
## Caffeine.consumption   9.504963e-05
## Awakenings            -1.161318e-02
## Exercise.frequency     3.403096e-03
## Alcohol.consumption   -4.546457e-03
## GenderMale             3.902944e-03
## Smoking.statusYes     -3.910745e-02
## y1                     1.209616e-01
```

Compare to OLS

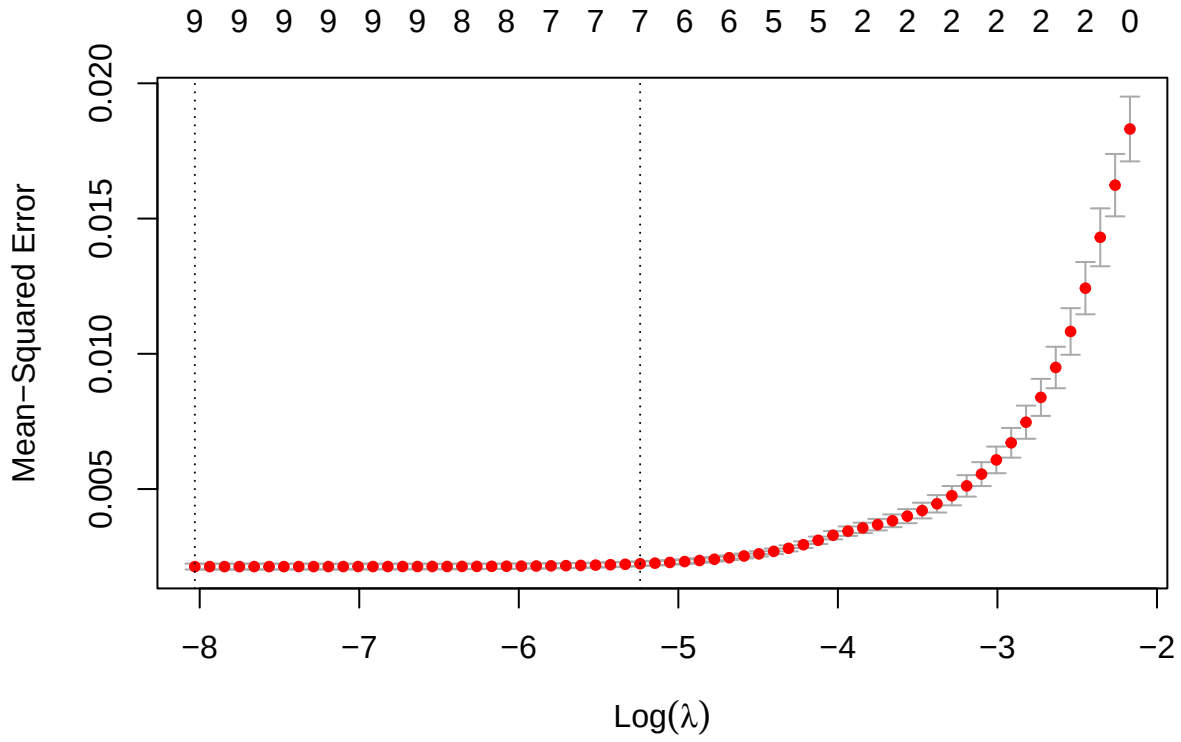
```
ols.regression <- lm(Sleep.efficiency ~ ., data = wse)
#OLS vs RIDGE
round(cbind(OLS = coef(ols.regression),
            ridge = c(ridge.model$a0,                                    #intercept for
                      as.vector(ridge.model$beta))),4)
```

```
##                           OLS   ridge
## (Intercept)            0.4022  0.4261
## Deep.sleep.percentage  0.0040  0.0038
## REM.sleep.percentage   0.0060  0.0055
## Caffeine.consumption   0.0001  0.0001
## Awakenings            -0.0105 -0.0116
## Exercise.frequency     0.0029  0.0034
## Alcohol.consumption   -0.0038 -0.0045
## GenderMale             0.0042  0.0039
## Smoking.statusYes     -0.0395 -0.0391
## y1                     0.1271  0.1210
```

LASSO Regression

```
#Penalty type (alpha=1 is lasso
#and alpha=0 is the ridge)
cv.lambda.lasso <- cv.glmnet(x=X, y=Y,
                             alpha = 1)
```

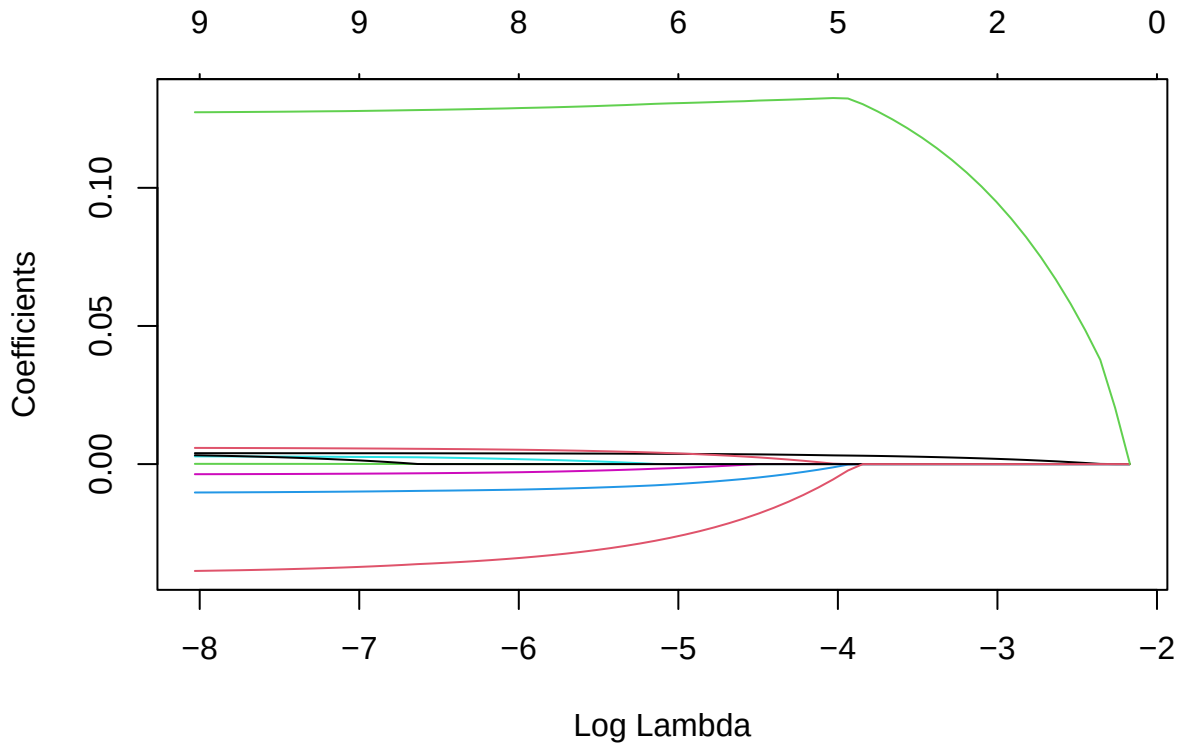
```
plot(cv.lambda.lasso)                          #MSE for several lambdas
```



```
cv.lambda.lasso
```

```
##
## Call:  cv.glmnet(x = X, y = Y, alpha = 1)
##
## Measure: Mean-Squared Error
##
##       Lambda Index  Measure        SE Nonzero
## min 0.000325    64 0.002133 1.082e-04       9
## 1se 0.005303    34 0.002237 8.568e-05       7
```

```
#Lasso path
plot(cv.lambda.lasso$glmnet.fit,
     "lambda", label=FALSE)
```



Final Model

```
l.lasso.min <- cv.lambda.lasso$lambda.min
l.lasso.min
```

```
## [1] 0.0003253697
```

```
lasso.model <- glmnet(x=X, y=Y,
                      alpha  = 1,
                      lambda = l.lasso.min)
lasso.model$beta
```

```
## 9 x 1 sparse Matrix of class "dgCMatrix"
##                                    s0
## Deep.sleep.percentage  3.946888e-03
## REM.sleep.percentage   5.852472e-03
## Caffeine.consumption   8.256676e-05
## Awakenings            -1.028367e-02
## Exercise.frequency     2.793621e-03
## Alcohol.consumption   -3.667998e-03
## GenderMale             3.201197e-03
## Smoking.statusYes     -3.868242e-02
## y1                     1.273402e-01
```